

Final Paper

```
library(tidyverse)
library(lubridate)
library(broom)
library(glue)
library(patchwork)
library(scales)

theme_set(
  theme_minimal(base_size = 13) +
    theme(
      plot.title = element_text(face = "bold"),
      plot.subtitle = element_text(size = 11),
      legend.position = "bottom",
      panel.grid.minor = element_blank()
    )
)

# -----
# Load data
# -----
gdelt <- read_csv("gdelt_gkg_immigration_monthly.csv", show_col_types = FALSE)
reddit_llm <- read_csv("reddit_llm_monthly.csv", show_col_types = FALSE)
ice <- read_csv("ice_monthly.csv", show_col_types = FALSE)
cbp <- read_csv("cbp_monthly.csv", show_col_types = FALSE)

# -----
# Clean dates
# -----
gdelt <- gdelt %>% mutate(month = as.Date(month))
reddit_llm <- reddit_llm %>% mutate(month = as.Date(month))
ice <- ice %>% mutate(month = as.Date(month))
cbp <- cbp %>% mutate(month = as.Date(month))
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# -----
# Merge and standardize
# -----
analysis_df <- gdelt %>%
  full_join(reddit_llm, by = "month") %>%
  full_join(ice, by = "month") %>%
  full_join(cbp, by = "month") %>%
  arrange(month) %>%
  mutate(
    news_z = as.numeric(scale(n_articles)),
    tone_z = as.numeric(scale(mean_tone)),
    arrests_z = as.numeric(scale(arrests)),
    cbp_z = as.numeric(scale(encounters)),
    sentiment_z = as.numeric(scale(avg_sentiment)),
    reddit_posts_z = as.numeric(scale(n_posts)),
    divergence_cbp = news_z - cbp_z,
    divergence_ice = news_z - arrests_z,
    divergence_sentiment_cbp = sentiment_z - cbp_z
  )

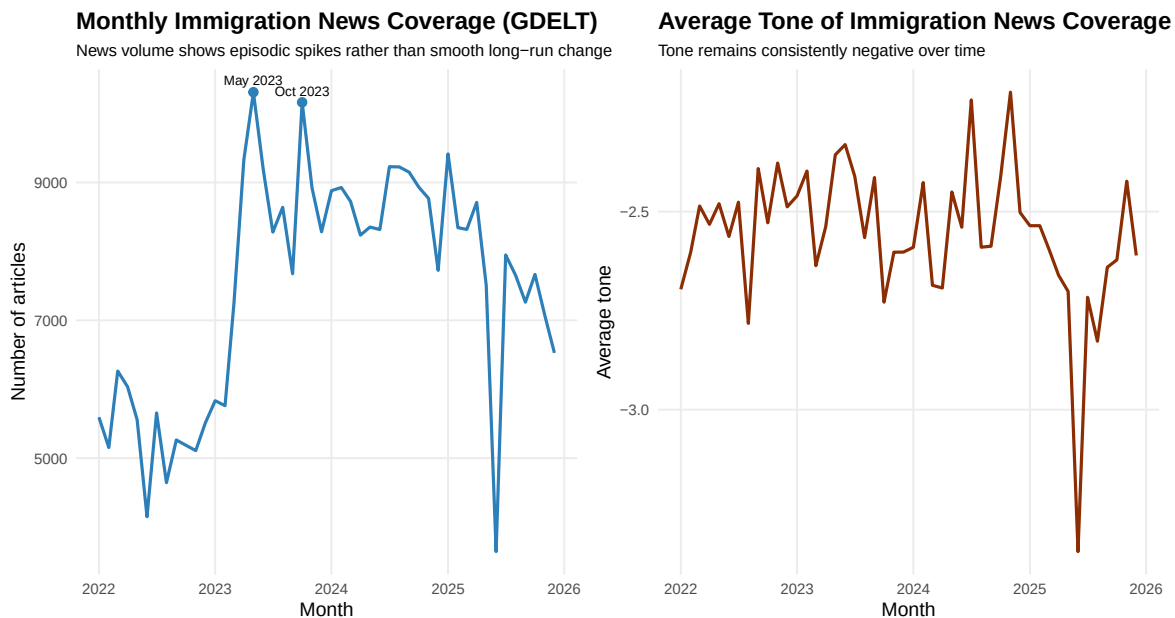
# -----
# 1. GDELT news volume
# -----
p1 <- ggplot(gdelt, aes(month, n_articles)) +
  geom_line(linewidth = 1, color = "#2C7FB8") +
  geom_point(
    data = gdelt %>% slice_max(n_articles, n = 2),
    size = 2.5, color = "#2C7FB8"
  ) +
  geom_text(
    data = gdelt %>% slice_max(n_articles, n = 2),
    aes(label = format(month, "%b %Y")),
    vjust = -0.7, size = 3.2
  ) +
  labs(
    title = "Monthly Immigration News Coverage (GDELT)",
    subtitle = "News volume shows episodic spikes rather than smooth long-run change",
    x = "Month", y = "Number of articles"
  )

# -----
# 2. GDELT tone

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# -----
p2 <- ggplot(gdelt, aes(month, mean_tone)) +
  geom_line(linewidth = 1, color = "#8C2D04") +
  labs(
    title = "Average Tone of Immigration News Coverage",
    subtitle = "Tone remains consistently negative over time",
    x = "Month", y = "Average tone"
  )

print(p1 + p2)
```



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# -----
# 3. ICE and CBP plots
# -----
enforcement_long <- bind_rows(
  ice %>% transmute(month, value = arrests, series = "ICE administrative arrests"),
  cbp %>% transmute(month, value = encounters, series = "CBP encounters")
)

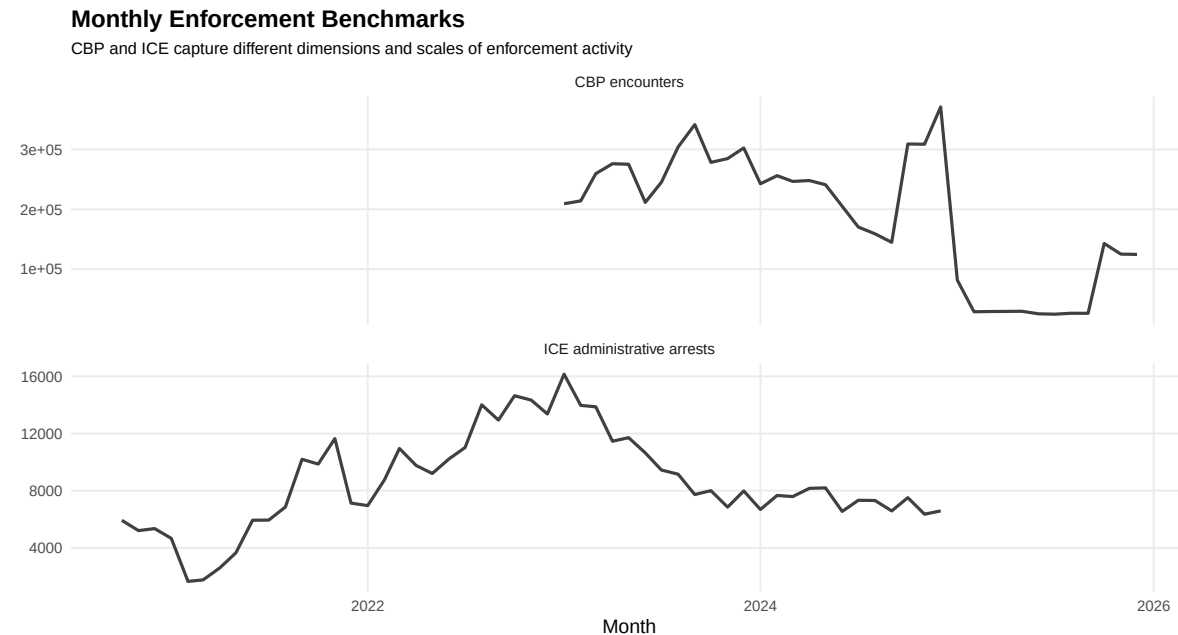
p3 <- ggplot(enforcement_long, aes(month, value)) +
  geom_line(linewidth = 1, color = "#404040") +
  facet_wrap(~series, scales = "free_y", ncol = 1) +
  labs(
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    title = "Monthly Enforcement Benchmarks",
    subtitle = "CBP and ICE capture different dimensions and scales of enforcement activity"
    x = "Month", y = NULL
  )

print(p3)

```



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# -----
# 4. Standardized comparison
# -----
std_plot_df <- analysis_df %>%
  select(month, news_z, cbp_z, arrests_z) %>%
  pivot_longer(-month, names_to = "series", values_to = "value") %>%
  mutate(
    series = recode(
      series,
      news_z = "News Coverage (GDELT)",
      cbp_z = "CBP Encounters",
      arrests_z = "ICE Arrests"
    )
  )

p4 <- ggplot(std_plot_df, aes(month, value, color = series)) +

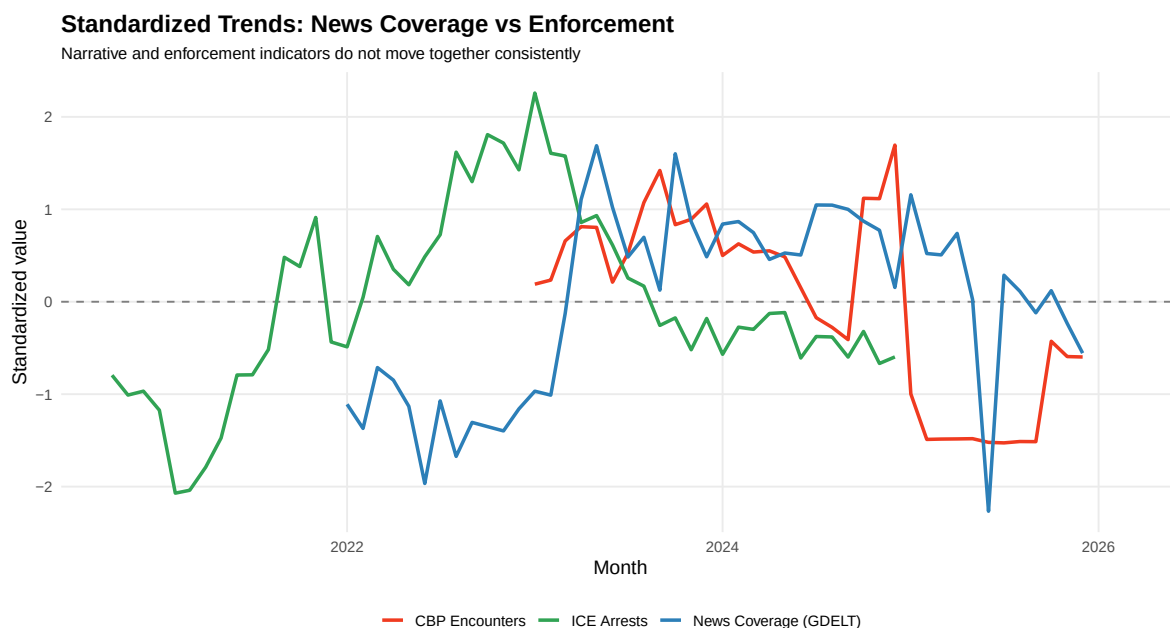
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geom_hline(yintercept = 0, linetype = "dashed", color = "gray50") +
geom_line(linewidth = 1.1, na.rm = TRUE) +
scale_color_manual(values = c(
  "News Coverage (GDELT)" = "#2C7FB8",
  "CBP Encounters" = "#F03B20",
  "ICE Arrests" = "#31A354"
)) +
labs(
  title = "Standardized Trends: News Coverage vs Enforcement",
  subtitle = "Narrative and enforcement indicators do not move together consistently",
  x = "Month", y = "Standardized value", color = NULL
)

print(p4)

```



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# -----
# 5. Divergence: news vs CBP
# -----
div_df <- analysis_df %>%
  select(month, divergence_cbp) %>%
  filter(!is.na(divergence_cbp))

p5 <- ggplot(div_df, aes(month, divergence_cbp)) +

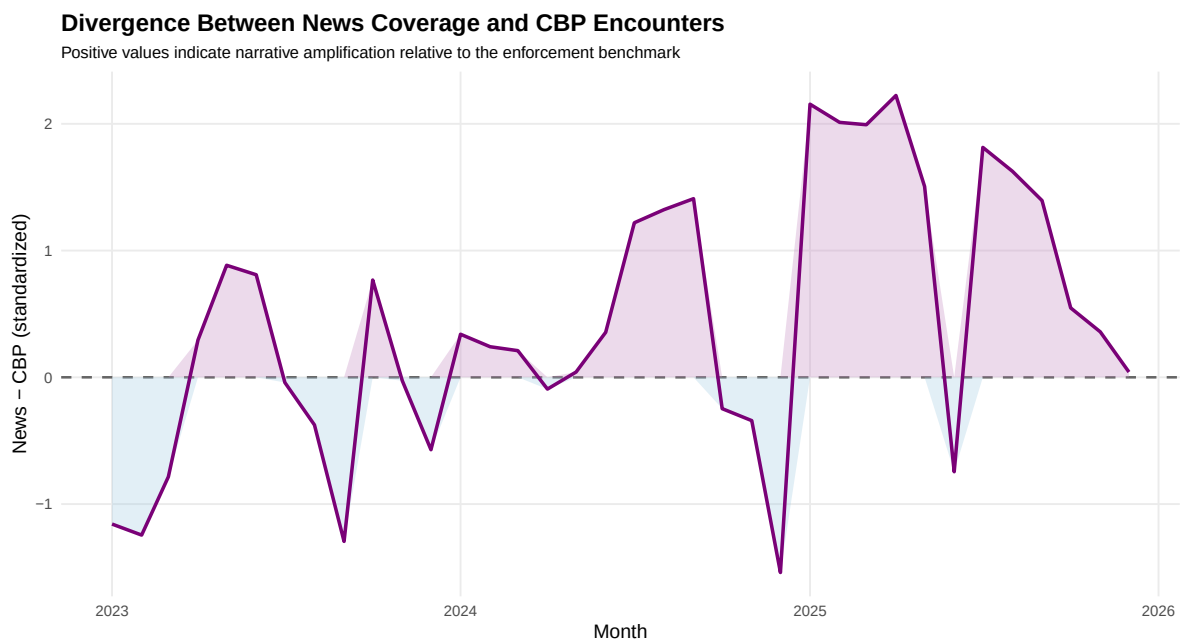
```

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geom_hline(yintercept = 0, linetype = "dashed", color = "gray40", linewidth = 0.8) +
geom_ribbon(
  aes(ymin = 0, ymax = pmax(divergence_cbp, 0)),
  fill = "#C994C7", alpha = 0.35
) +
geom_ribbon(
  aes(ymin = pmin(divergence_cbp, 0), ymax = 0),
  fill = "#9ECAE1", alpha = 0.30
) +
geom_line(color = "#7A0177", linewidth = 1.1) +
labs(
  title = "Divergence Between News Coverage and CBP Encounters",
  subtitle = "Positive values indicate narrative amplification relative to the enforcement",
  x = "Month", y = "News - CBP (standardized)"
)

print(p5)

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# -----
# 6. Regression: News -> CBP
# -----
reg_df <- analysis_df %>%
  select(news_z, cbp_z, month) %>%

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filter(!is.na(news_z), !is.na(cbp_z))

model_cbp <- lm(cbp_z ~ news_z, data = reg_df)

r2_txt <- round(glance(model_cbp)$r.squared, 3)
p_txt  <- round(glance(model_cbp)$p.value, 3)

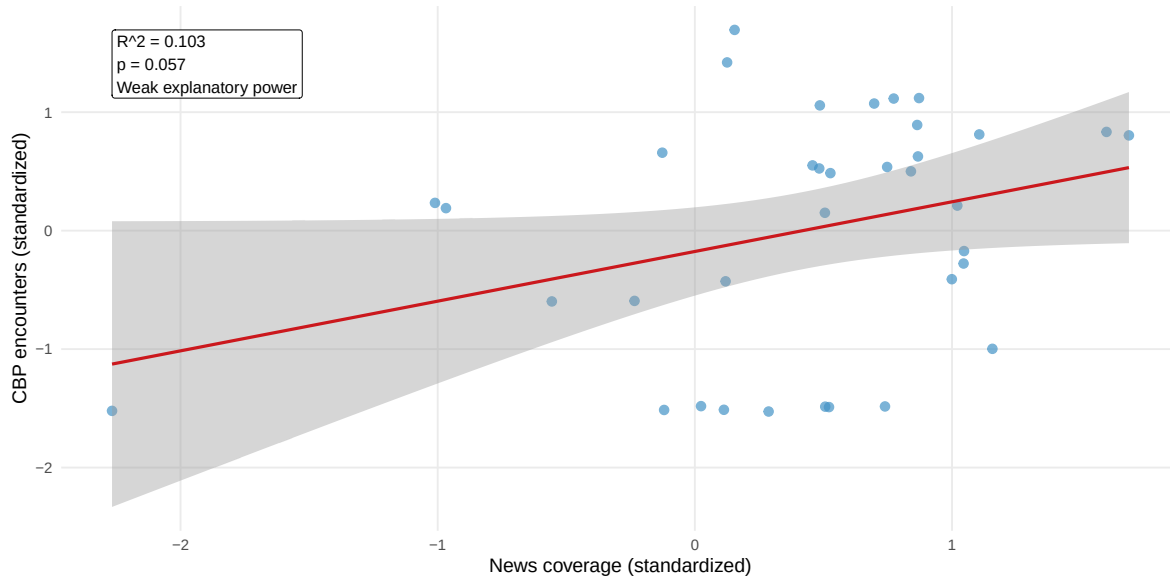
p6 <- ggplot(reg_df, aes(news_z, cbp_z)) +
  geom_point(alpha = 0.7, size = 2.5, color = "#4292C6") +
  geom_smooth(method = "lm", se = TRUE, color = "#CB181D", linewidth = 1) +
  annotate(
    "label",
    x = min(reg_df$news_z),
    y = max(reg_df$cbp_z),
    hjust = 0, vjust = 1,
    label = glue("R^2 = {r2_txt}\np = {p_txt}\nWeak explanatory power"),
    size = 3.8
  ) +
  labs(
    title = "Relationship Between News Coverage and CBP Encounters",
    subtitle = "News coverage is associated with CBP activity, but only weakly",
    x = "News coverage (standardized)",
    y = "CBP encounters (standardized)"
  )

print(p6)

```

Relationship Between News Coverage and CBP Encounters

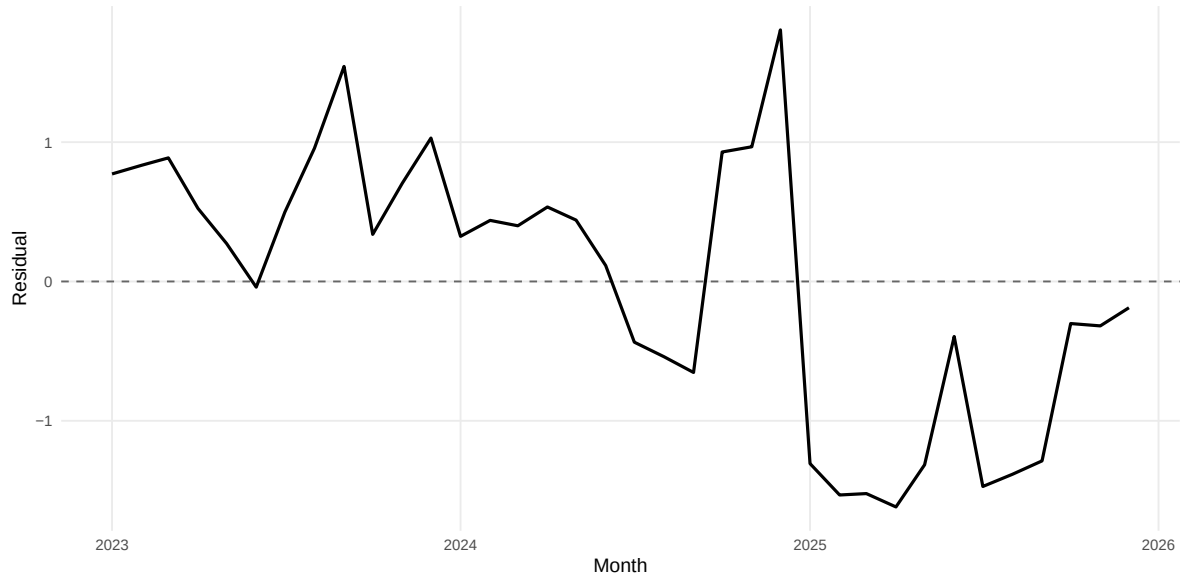
News coverage is associated with CBP activity, but only weakly



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# -----  
# 7. Residual mismatch  
# -----  
reg_df <- reg_df %>%  
  mutate(resid_cbp = residuals(model_cbp))  
  
p7 <- ggplot(reg_df, aes(month, resid_cbp)) +  
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray40") +  
  geom_line(color = "black", linewidth = 1) +  
  labs(  
    title = "Residual Mismatch: News Predicting CBP Encounters",  
    subtitle = "Large residuals indicate months where narrative measures fail to track the b",  
    x = "Month", y = "Residual"  
  )  
  
print(p7)
```

Residual Mismatch: News Predicting CBP Encounters

Large residuals indicate months where narrative measures fail to track the benchmark



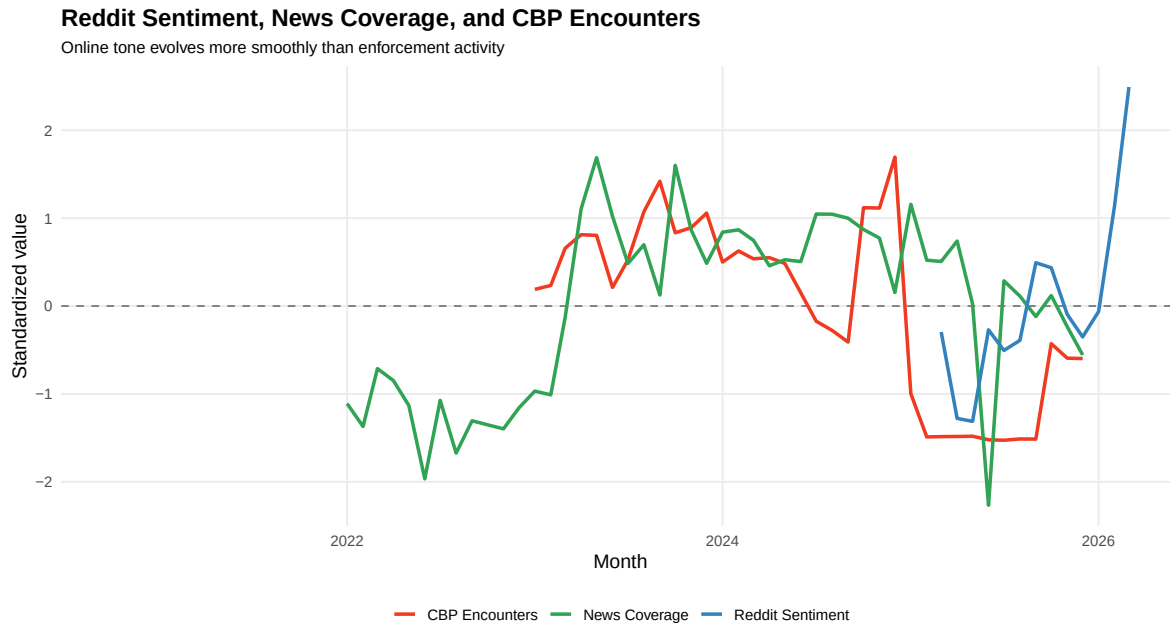
```
# -----  
# 8. Reddit sentiment comparison  
# -----  
sent_compare_df <- analysis_df %>%  
  select(month, sentiment_z, news_z, cbp_z) %>%  
  pivot_longer(-month, names_to = "series", values_to = "value") %>%  
  mutate(  
    series = recode(  
      series,  
      sentiment_z = "Reddit Sentiment",  
      news_z = "News Coverage",  
      cbp_z = "CBP Encounters"  
    )  
  )  
  
p8 <- ggplot(sent_compare_df, aes(month, value, color = series)) +  
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray50") +  
  geom_line(linewidth = 1.1, na.rm = TRUE) +  
  scale_color_manual(values = c(  
    "Reddit Sentiment" = "#3182BD",  
    "News Coverage" = "#31A354",  
    "CBP Encounters" = "#F03B20"  
  )) +  
  labs(  
    title = "Residual Mismatch: News Predicting CBP Encounters",  
    subtitle = "Large residuals indicate months where narrative measures fail to track the benchmark",  
    xlab = "Month",  
    ylab = "Residual",  
    legend = "Series"  
  )
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    title = "Reddit Sentiment, News Coverage, and CBP Encounters",
    subtitle = "Online tone evolves more smoothly than enforcement activity",
    x = "Month", y = "Standardized value", color = NULL
  )

print(p8)

```



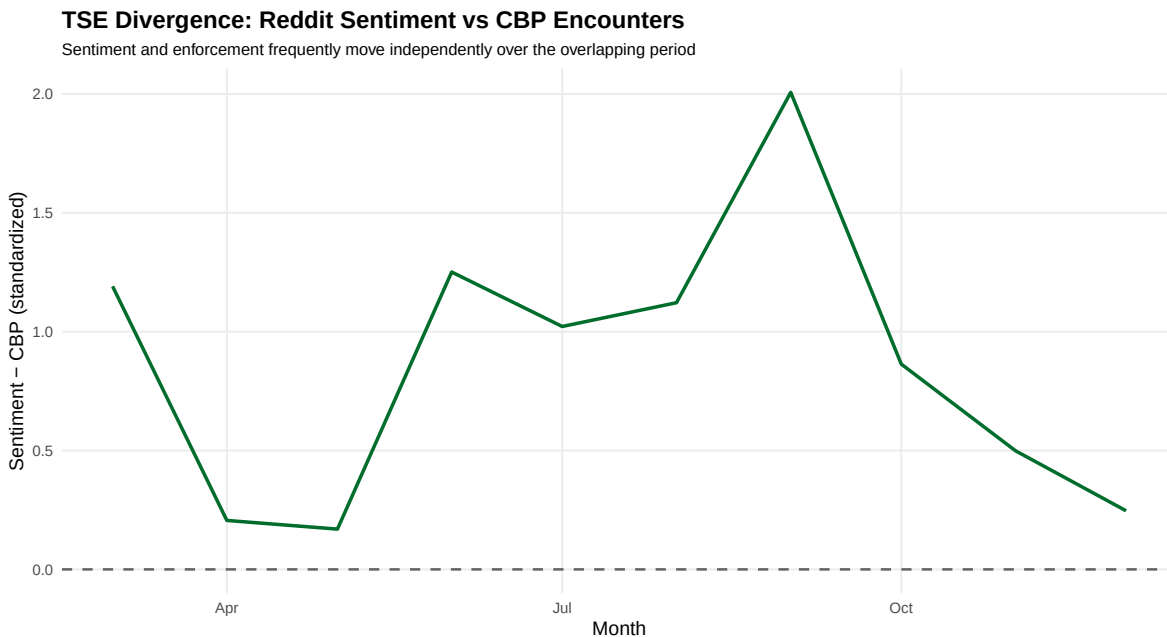
```

# -----
# 9. Sentiment divergence
# -----
sent_div_df <- analysis_df %>%
  select(month, divergence_sentiment_cbp) %>%
  filter(!is.na(divergence_sentiment_cbp))

p9 <- ggplot(sent_div_df, aes(month, divergence_sentiment_cbp)) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray40", linewidth = 0.8) +
  geom_line(color = "#006D2C", linewidth = 1.1) +
  labs(
    title = "TSE Divergence: Reddit Sentiment vs CBP Encounters",
    subtitle = "Sentiment and enforcement frequently move independently over the overlapping",
    x = "Month", y = "Sentiment - CBP (standardized)"
  )

```

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print(p9)
```



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# -----  
# 10. Regression: Sentiment -> CBP  
# -----  
sent_reg_df <- analysis_df %>%  
  select(sentiment_z, cbp_z) %>%  
  filter(!is.na(sentiment_z), !is.na(cbp_z))  
  
model_sentiment_cbp <- lm(cbp_z ~ sentiment_z, data = sent_reg_df)  
  
sent_r2 <- round(glance(model_sentiment_cbp)$r.squared, 3)  
sent_p <- round(glance(model_sentiment_cbp)$p.value, 3)  
  
p10 <- ggplot(sent_reg_df, aes(sentiment_z, cbp_z)) +  
  geom_point(alpha = 0.7, size = 2.5, color = "#74C476") +  
  geom_smooth(method = "lm", se = TRUE, color = "#E6550D", linewidth = 1) +  
  annotate(  
    "label",  
    x = min(sent_reg_df$sentiment_z),  
    y = max(sent_reg_df$cbp_z),  
    hjust = 0, vjust = 1,
```

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    label = glue("R^2 = {sent_r2}\\np = {sent_p}\\nNon-significant association"),
    size = 3.8
  ) +
  labs(
    title = "Relationship Between Reddit Sentiment and CBP Encounters",
    subtitle = "Sentiment captures some variation in the narrative environment, but is not a",
    x = "Reddit sentiment (standardized)",
    y = "CBP encounters (standardized)"
  )

print(p10)

```

